Retrospective Tumor Microenvironment Analysis from a Personalized neoTCR-T Cell Therapy Clinical Trial and Its Potential Applications for Off-the-Shelf HPV TCR T Cell Therapy

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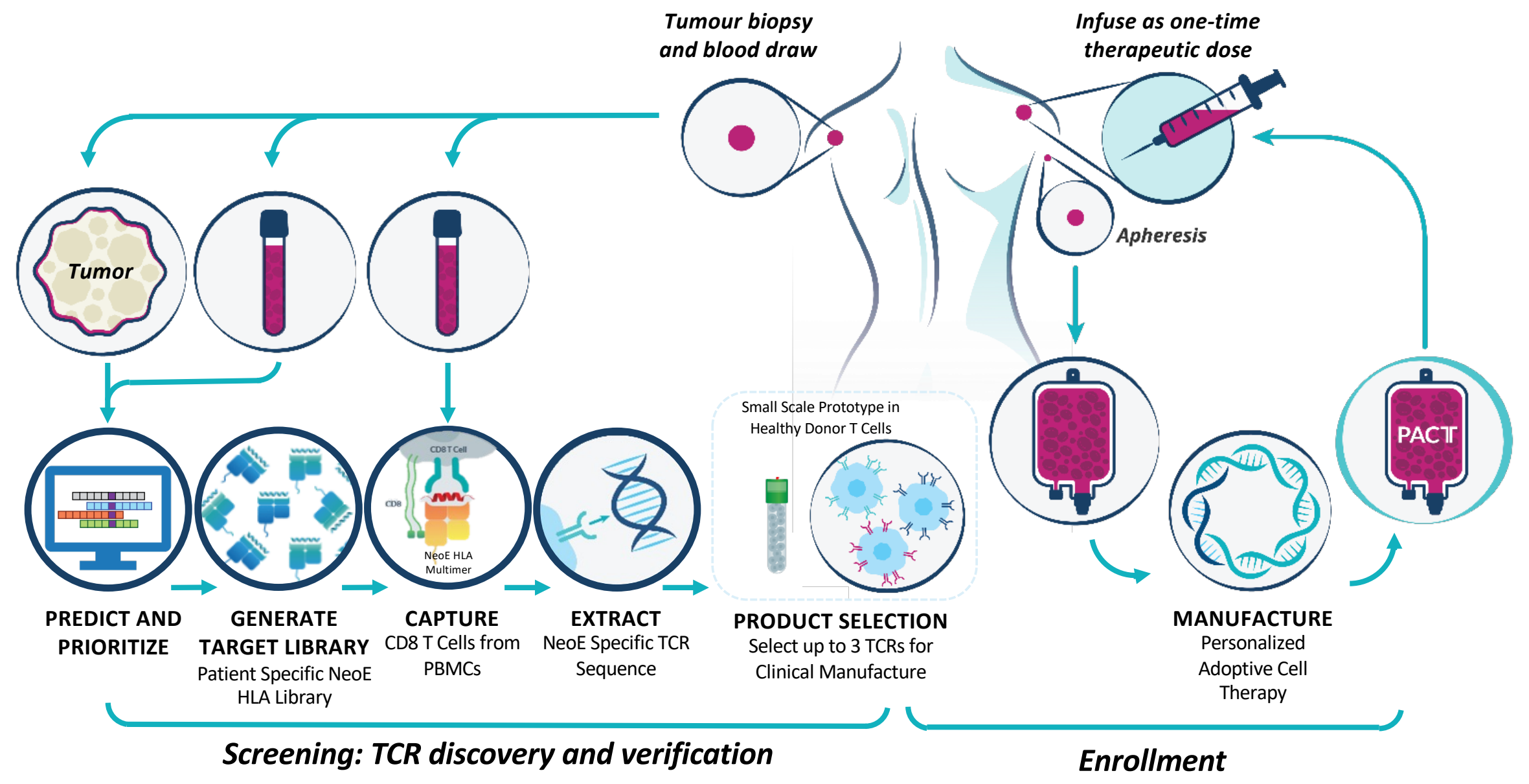
Abstract #712

Background

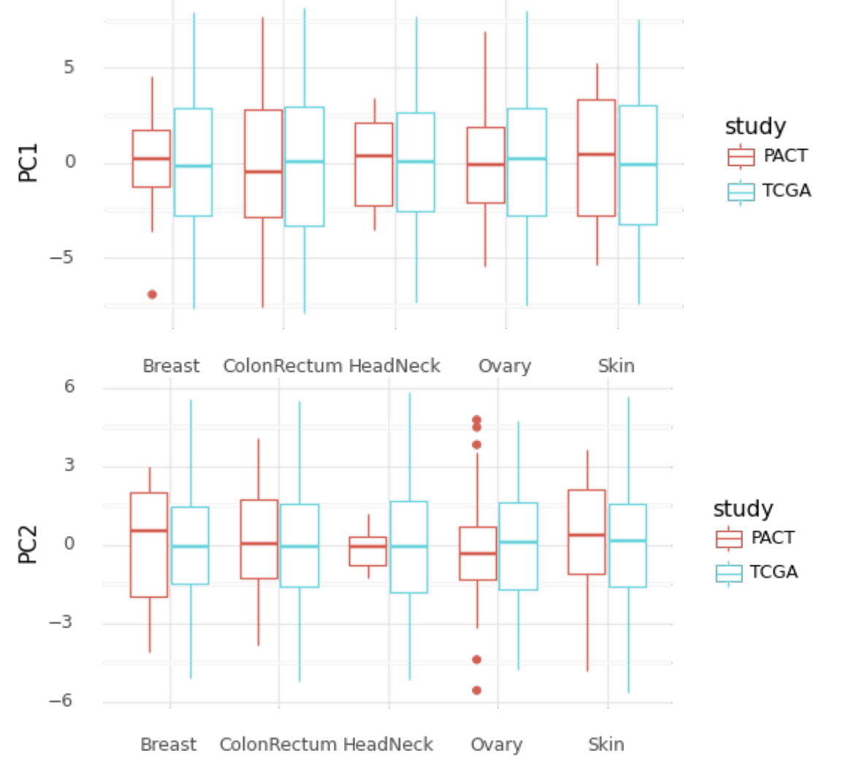
We have developed a state-of-the-art approach to validate predicted neoepitopes (neoEs) and their cognate T cell receptors (**neoTCR**s) by capturing neoepitope-specific T cells from peripheral blood. This neoTCR discovery and validation process is being applied in a phase I clinical trial (**NCT03970382**) evaluating personalized neoTCR-T cell therapy to treat patients across eight solid tumor types (PACT-0101). We have since expanded our mission to discover off-the-shelf TCRs against recurrent epitopes including HPV+ cancers. The tumor microenvironment (**TME**) has been shown to critically influence responses to immunotherapy making it paramount to explore in the context of adoptive cell therapy. Using four previously described immune signature TME subtypes, we performed retrospective analysis of our patients, correlated the signatures with TCR capture, and classified potential future patient populations using TCGA data.


Tumour biopsy and blood draw
Infuse as one-time therapeutic dose
Tumor
Apheresis
Small Scale Prototype in Healthy Donor T Cells
CD8 T Cell
CD8
NeoE HLA Multimer
PACT
PREDICT AND PRIORITIZE
GENERATE TARGET LIBRARY Patient Specific NeoE HLA Library
CAPTURE CD8 T Cells from PBMCs
EXTRACT NeoE Specific TCR Sequence
PRODUCT SELECTION Select up to 3 TCRs for Clinical Manufacture
MANUFACTURE Personalized Adoptive Cell Therapy
Screening: TCR discovery and verification
Enrollment

Schematic of the workflow from screening (TCR discovery and verification) through enrollment (cell manufacture and infusion).

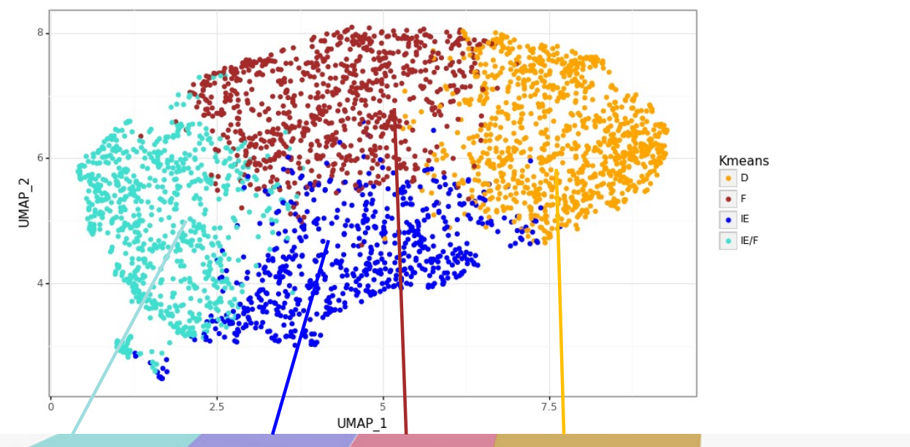
Methods

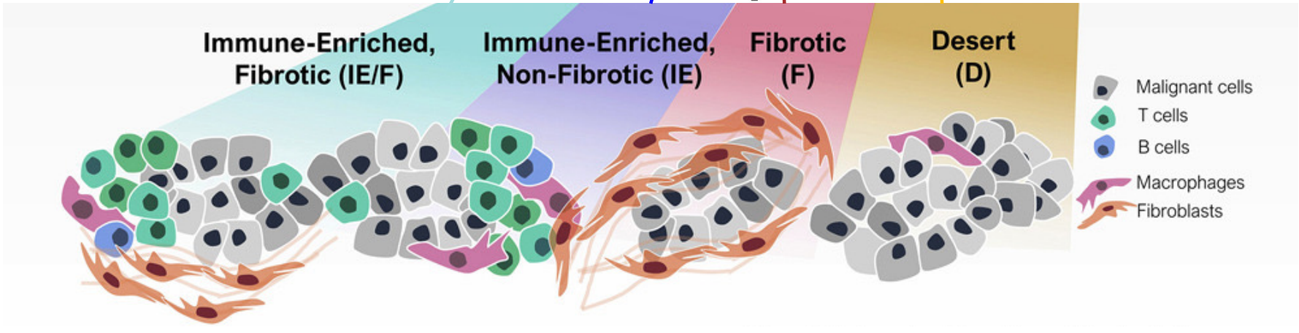
Using bulk RNA-seq data from biopsies, we examined the TME of both our clinical cohort (n=137) as well as TCGA H&N samples (n=485). To do this we measured **29 gene expression signatures** that capture specific aspects of different tumor infiltrates that were previously described as predictive to response to immunotherapy[1]. These signatures were used for K-means clustering with 4 distinct subsets, Immune enriched (**IE**), Depleted (**D**), Fibrotic (**F**), and Immune enriched/Fibrotic (**IE/F**) across 6 different cancer types.



After batch correction and ssGSEA calculations, PACT and TCGA cohorts were quantile transformed. A principal component analysis after transformations shows a similar distribution of PCs scores between PACT and TCGA, enabling direct comparison between the two groups and adding power for future predictions.

K-means clustering was used to assign a TME subtype for each sample.



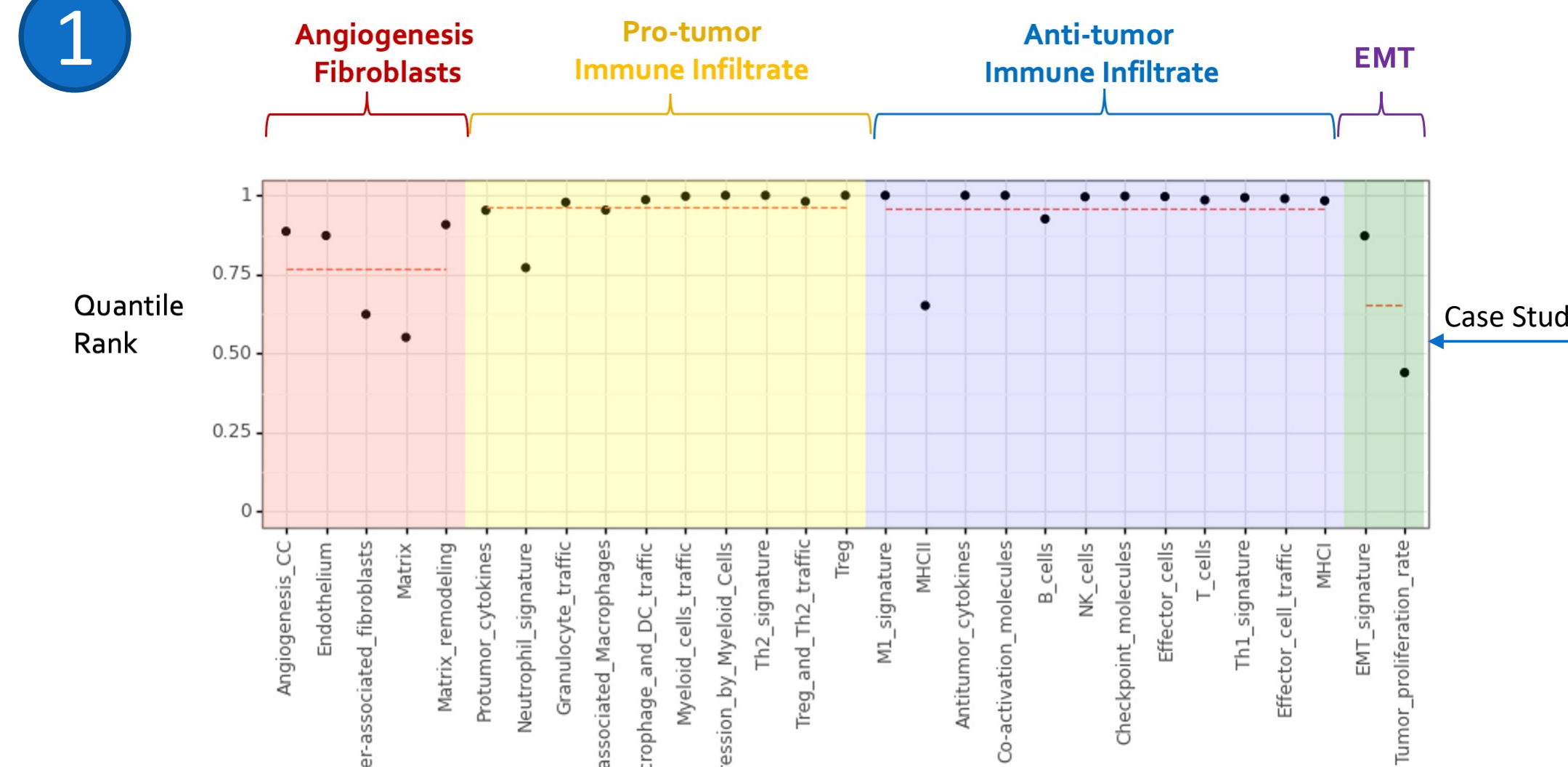


Adopted from Bagaev et al., 2021, Cancer Cell 39, 845–865

Results

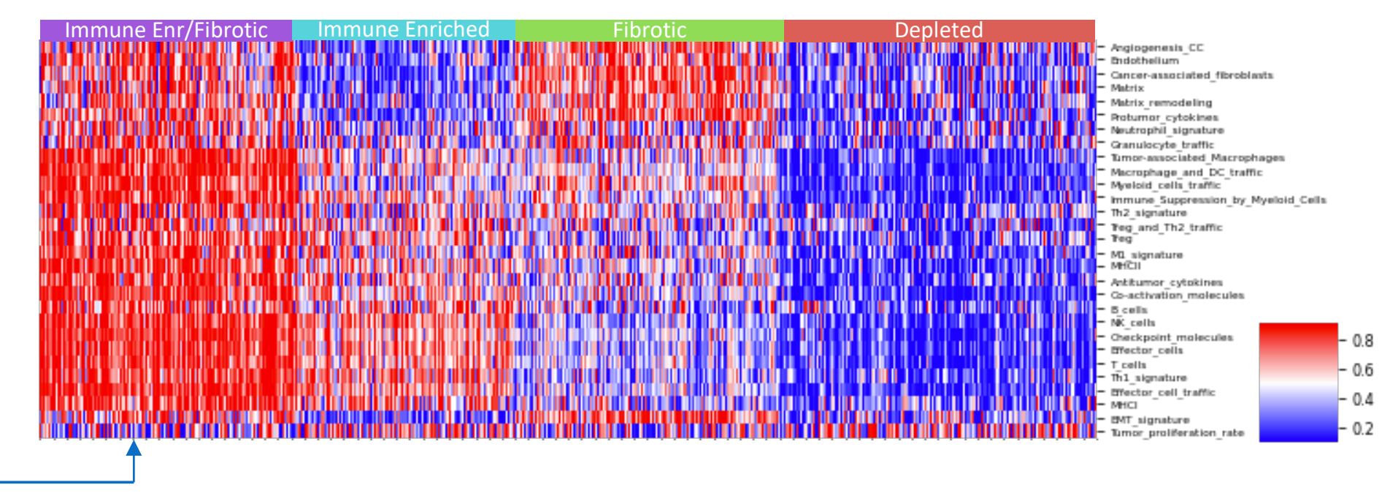
1 TME Classification of PACT-0101 Clinical Trial Screening Samples and TCGA Samples

Example of a High Confidence IE/F Subtype Prediction



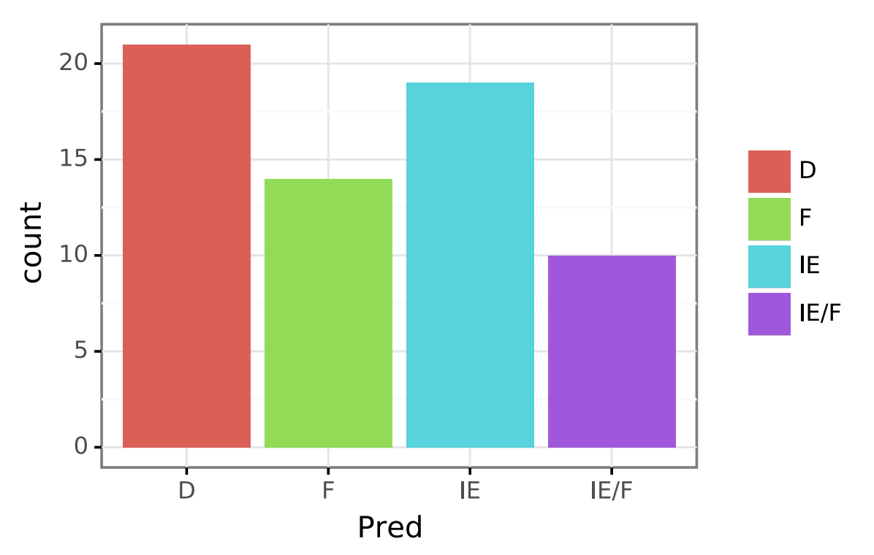
Quantile rank of ssGSEA scores for 4 major groups[1] of gene expression signatures (x-axis) for one sample. Dotted red lines indicate the average for each group.

Four Distinct Gene Expression Patterns Observed Across TCGA Samples



Heatmap of ssGSEA scores for pathways of interest across TCGA samples showing 4 distinct clusters labeled above. Scores for one sample case study are displayed in the figure on the left (follow the arrow).

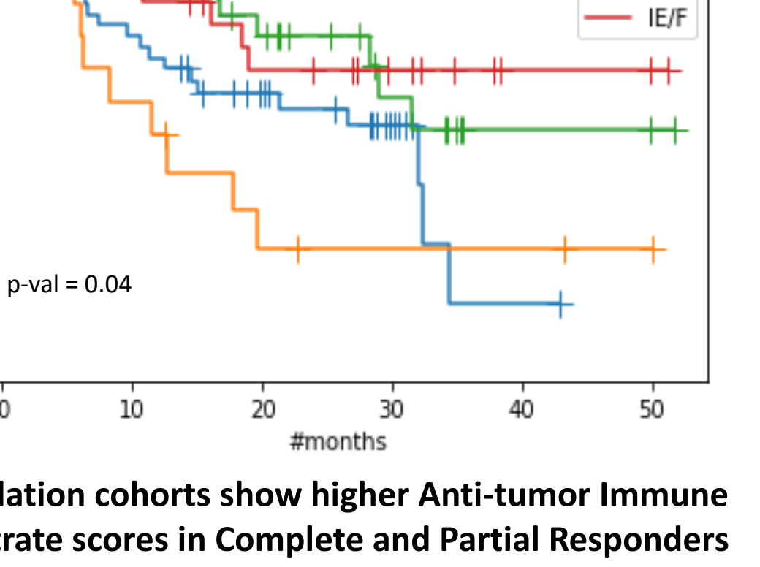
PACT TME Subtypes at Screening



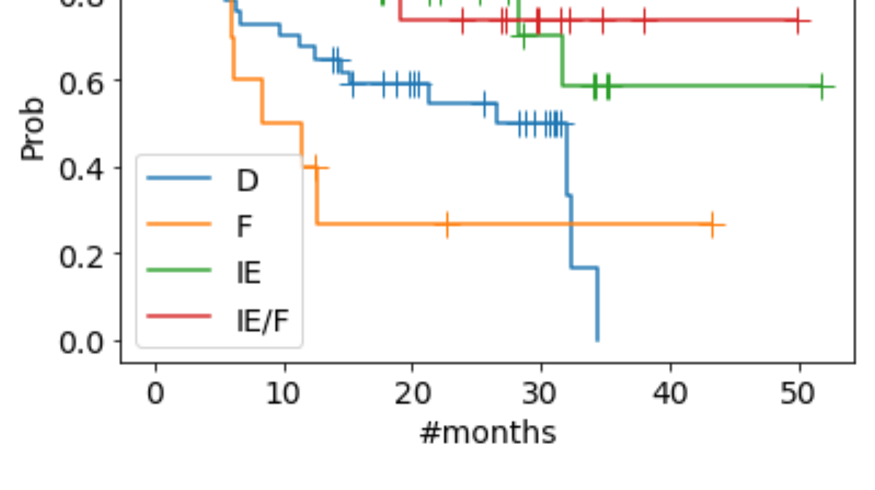
TME subtype classification of screening samples from the PACT-0101 clinical trial. Subtypes of dosed patients are below.

2 TME Classification Validation

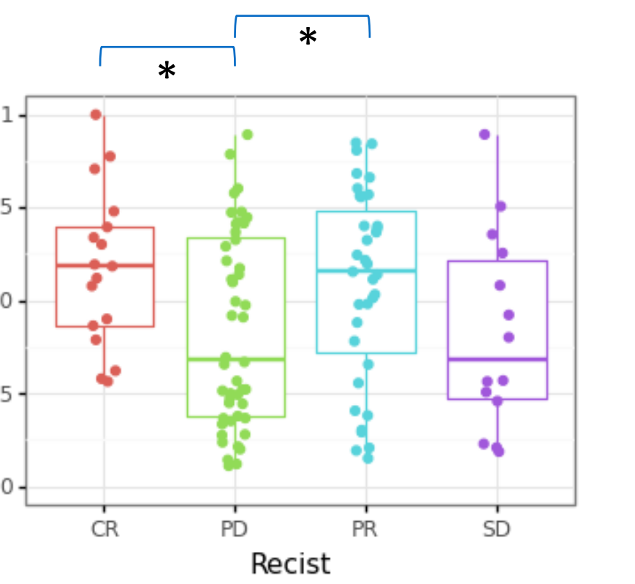
Predicted TME Subtypes are Predictive of Response to ICI



High Confidence Predictions Show Even Better Separation by Subtype



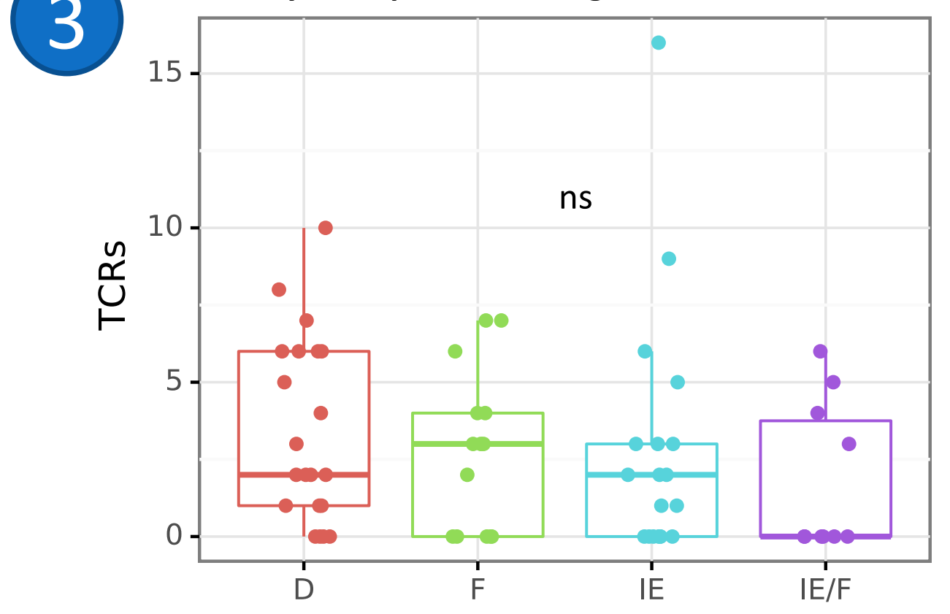
Validation cohorts show higher Anti-tumor Immune Infiltrate scores in Complete and Partial Responders relative to progressive disease patients



(Top left) Overall survival plot analysis of 3 ICI cohorts stratified by TME subtype for melanoma samples previously described[1]. IE and IE/F both do better than D and F subtypes. (**Top right)** Same plot but with high confidence TME subtype samples (Probability > 0.8). (**Bottom left**) Aggregated anti-tumor immune infiltrate ssGSEA scores stratified by response to ICI in prospective validation cohorts.

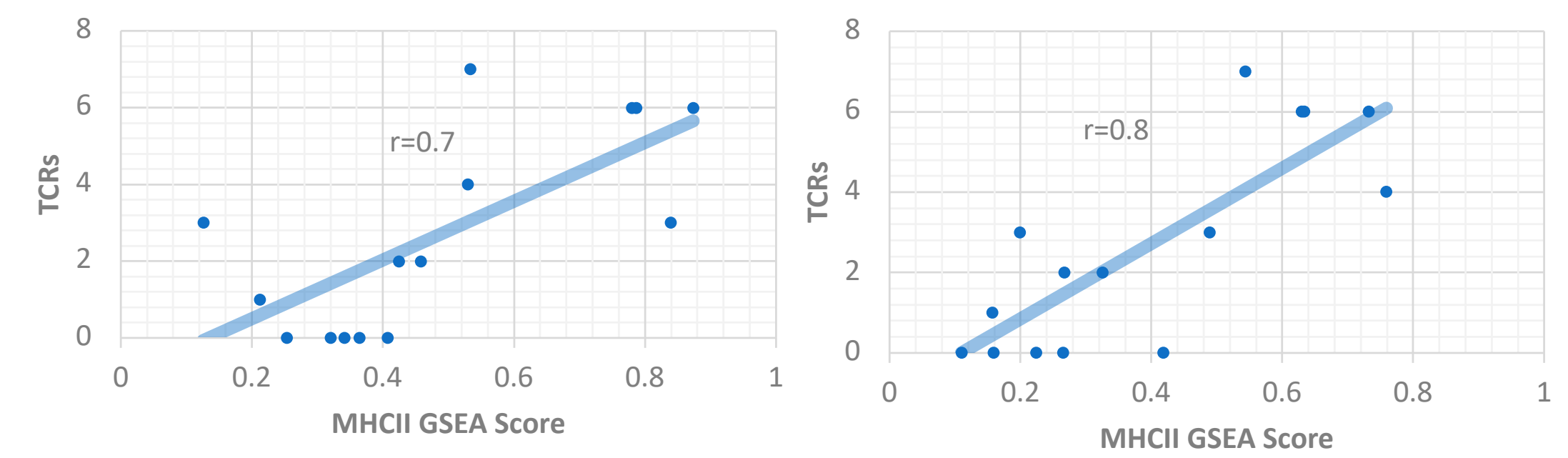
3 TME Correlates with Autologous TCR Capture

TME Subtype Does Not Significantly Influence Ability to Capture Autologous TCRs



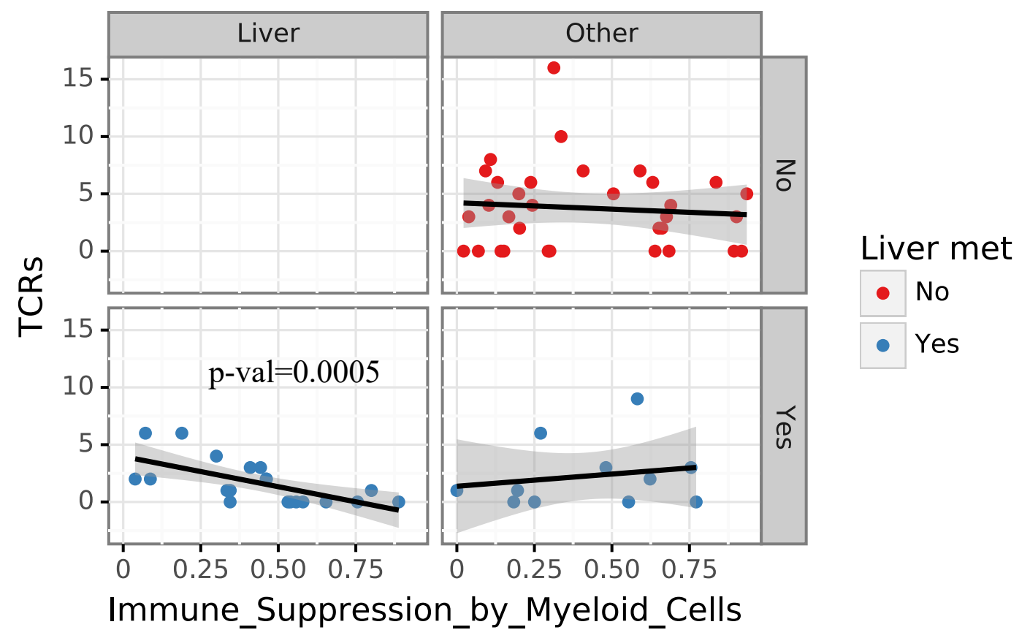
Isolated autologous TCRs stratified by predicted TME subtype.

Autologous TCR Capture is Positively Correlated with TREG/TH2 Trafficking I and MHCII in Ovarian Cancers at Screening



Correlation of autologous TCRs captured versus individual gene expression signatures for (left) Treg and Th2 Traffic (p<0.004) and (right) MHCII for ovarian cancers (p<0.001).

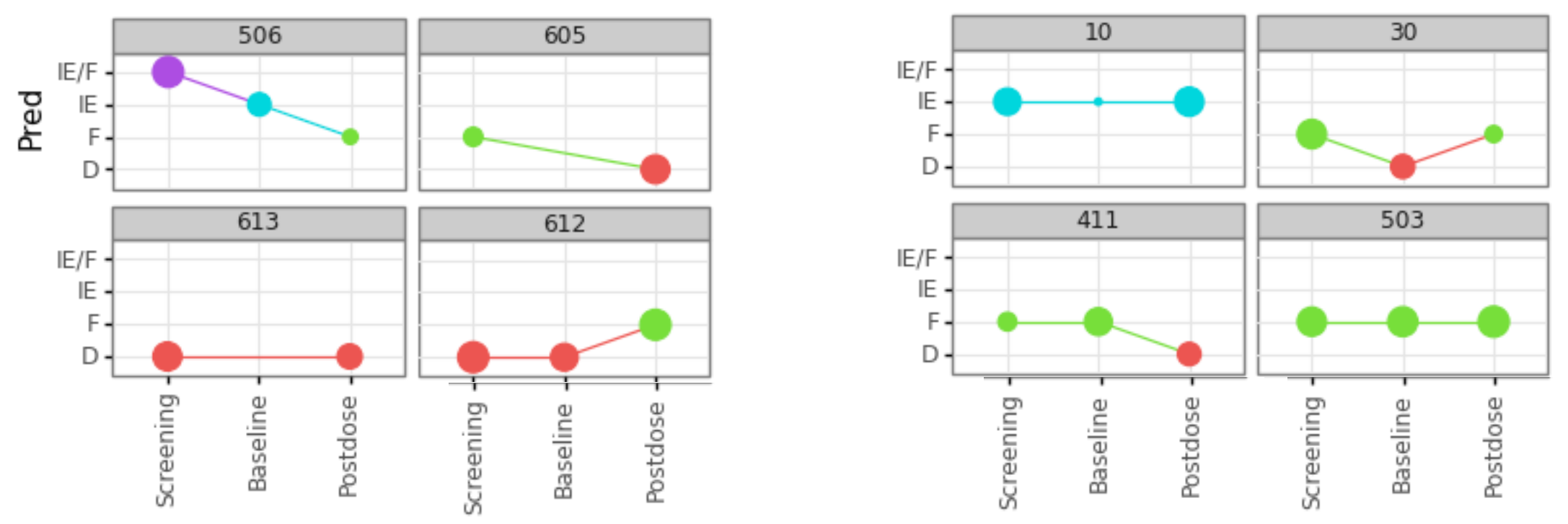
Autologous TCR Capture is Negatively Correlated with Immune Suppression by Myeloid Cells in Liver Samples.



Number of autologous TCRs captured vs. the immune suppression by myeloid cells gene expression signature stratified by whether the sample had a liver met (rows) and the location of biopsy (columns).

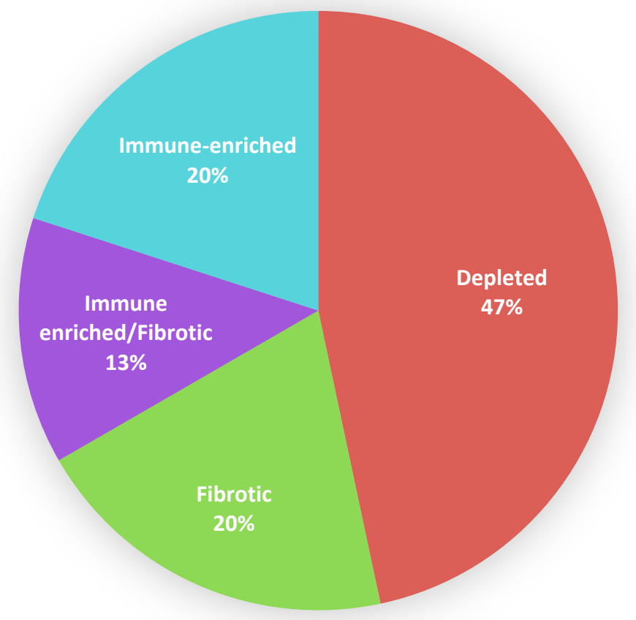
4 TME Characteristics of Longitudinal Samples and Dosed Individuals

Longitudinal Samples Rarely Show Switches Between Favorable (IE or IE/F) and Unfavorable Immune Subtypes (D or F)



Longitudinal sample TME classification (often spanning > 1 year) from archival screening, baseline and post-dosing biopsies, often taken from different locations. Numbers on top indicate patient ID. Only one patient (506) of eight went from favorable to unfavorable. Larger points represent higher predicted subtype probabilities.

67% of Dosed Patients Had an Unfavorable (D or F) TME Subtype

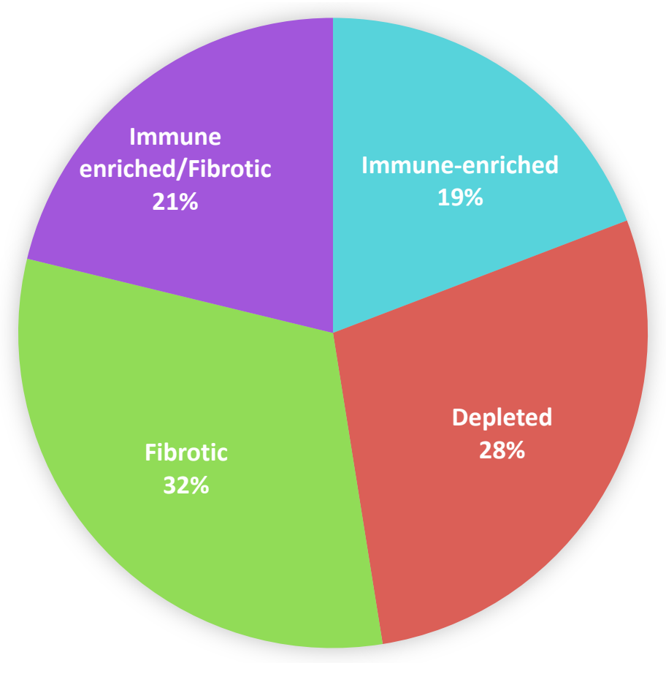


TME subtype of dosed individuals (N=15 patients). Clinical outcomes are described in Poster 1478[2].

5 TME Classification of Head and Neck Cancer

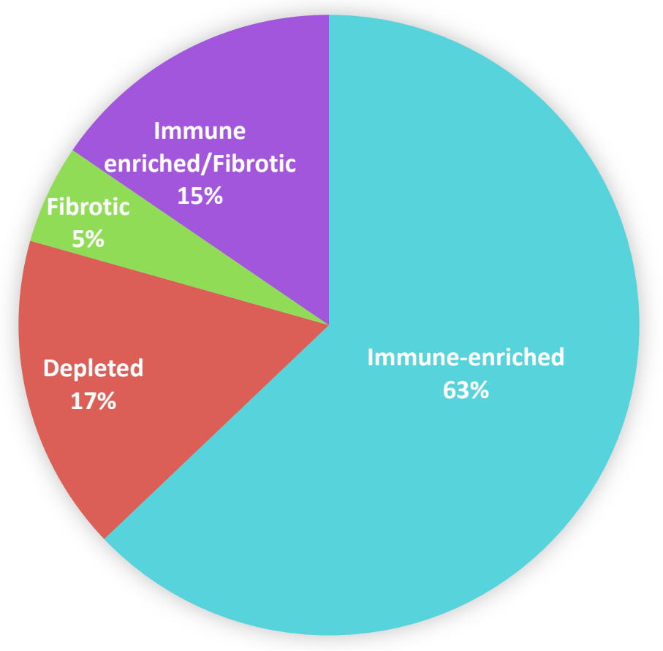
HPV+ Tumors are Enriched for Favorable (IE or IE/F) Immunotherapy TME Subtypes (78%)

HPV– (N = 414)



HPV- HNSC TCGA TME Subtypes

HPV+ (N = 71)



HPV+ HNSC TCGA TME Subtypes

Conclusions

- A pan-cancer TME subtype classifier was developed to classify patient tumor biopsies in a neoantigen neoTCR clinical trial.
- Retrospective analysis shows no correlation with TME subtype and our ability to isolate autologous TCRs from those patients.
- TME subtypes of dosed patients were predominately unfavorable for immunotherapy.
- In contrast, HPV+ cancers are enriched for immunotherapy responsive subtypes, suggesting it may be a favorable cancer indication for adoptive cell therapy.

Acknowledgements and Refs

Thank you to the Patients and their Families

1.) Bagaev A., …, Fowler N. Conserved pan-cancer microenvironment subtypes predict response to immunotherapy. Cancer Cell. 2021 Jun 14;39(6):845-865.e7. doi: 10.1016/j.ccell.2021.04.014. Epub 2021 May 20. PMID: 34019806.
2.) Foy S., A phase I study of personalized adoptive TCR T cell therapy in patients with solid tumors: safety, efficacy, and T cell trafficking to tumors of non-virally gene edited T cells, Poster 1478, SITC 2022

Trial Registration: NCT03970382
Ethics Approval Statement: This study was approved by the review boards at each clinical site enrolling patients
This study was funded by PACT Pharma.